P525/1

CHEMISTRY

JULY/August, 2026

2 ¾ Hours

MUKONO EXAMINATION COUNCIL 2026

Uganda Advanced Certificate of Education

CHEMISTRY

PAPER 1

TIME: 2 HOURS 45 MINUTES

INSTRUCTIONS TO CANDIDATES

*This examination paper consists of two sections; **A** and **B.** It has six examination items.*

*Section **A** has **two** compulsory items.*

*Section **B** has two parts; **I** and **II.** Respond to only one item from each part.*

*Respond to **four** items in all.*

*Any additional item(s) responded to will **not** be scored.*

*All responses **must** be written in the answer booklet(s) provided.*

Begin each item on a fresh page.

SECTION A

Respond to **both** items from this section

1.

A group of young science innovators in Jinja is developing low-cost cold packs for sports injuries and first-aid use in schools. They decide to use sodium thiosulphate, $Na_2S_2O_3$, because its dissolution in water absorbs heat and can therefore produce a cooling effect.

During testing, 12.64 g of solid sodium thiosulphate was added to 100.0 cm^3 of water in an insulated cup. The temperature fell from 25.0 °C to 21.4 °C. The trainees were unable to use the temperature change to determine the enthalpy of solution and lattice enthalpy of sodium thiosulphate, or to comment on the stability of its ionic lattice.

After using the cold packs, the team became concerned about disposing of thiosulphate solution into the environment. Their chemistry mentor advised them to recover sulphur by reacting the used sodium thiosulphate solution with dilute hydrochloric acid:

$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$

The trainees also obtained rate data for sulphur formation but could not determine the orders of reaction, the rate equation, the effect of changing reactant concentrations, or the environmental and operational conditions that should be considered during sulphur recovery.

The following data were obtained at constant temperature.

Experiment	**$[Na_2S_2O_3]$ / mol dm^{-3}**	**[HCl] / mol dm^{-3}**	**Initial rate of sulphur formation / mol dm^{-3} s^{-1}**
1	0.020	0.100	2.40×10^{-4}
2	0.040	0.100	4.80×10^{-4}
3	0.020	0.200	4.80×10^{-4}

Data:

Molar mass of $Na_2S_2O_3$ = 158 g mol^{-1}
Density of water = 1.0 g cm^{-3}
Specific heat capacity of the resulting solution = 4.18 J g^{-1} K^{-1}
Total hydration enthalpy of the ions from one mole of $Na_2S_2O_3$ = -790 kJ mol^{-1}

Tasks

a) Analyze the thermochemical information provided and use it to comment on the suitability of sodium thiosulphate for the proposed cold packs.

b) Analyze and interpret the rate data obtained during the recovery of Sulphur.

c) Make appropriate recommendations to the trainees concerning the operation and environmental management of the processes described.

2.

A medicinal chemist working with a natural-products research laboratory in Uganda is investigating a plant traditionally used for treating minor skin infections. A volatile organic compound, X, has been isolated from the crushed leaves and is suspected to be the active ingredient.

Analysis showed that X contains carbon, hydrogen and oxygen only. Complete combustion of 0.296 g of X produced 0.857 g of carbon dioxide and 0.280 g of water. During a separation experiment at atmospheric pressure, a mixture of X and water boiled at 96 °C. At this temperature, the vapour pressure of water was 658 mmHg, while atmospheric pressure was 760 mmHg. The distillate contained 56.7% by mass of X. Use C = 12, H = 1, O = 16 and molar mass of water = 18 g mol^{-1}.

Compound X is immiscible with water and volatile, but it decomposes when heated strongly near its normal boiling point of 110 °C. The laboratory assistant needs to determine its formula, understand why its mixture with water boils below the boiling point of either pure liquid, and select a suitable method for isolating it from the plant material.

Tasks

a) Analyze the combustion data and establish the relevant chemical information about compound X.

b) Interpret the vapour-pressure and distillate-composition data in relation to the identity and behaviour of X.

c) Prepare a write-up to help the laboratory assistant isolate X and understand the practical importance of the method used.

SECTION B

PART I

*Respond to any **one** item in this part*

3.

Scientists from the National Nuclear Research Centre are monitoring a sealed radioactive source used to sterilise medical equipment in a regional referral hospital.

The source was installed several years earlier and emits ionising radiation continuously. Because radioactive nuclei decay spontaneously and unpredictably, the hospital management wants to determine whether the source still has sufficient activity for effective sterilization.

A team of Senior Six learners on industrial training was invited to analyse the source before a decision could be made about replacing it. The hospital physicist explained that radioactive decay transforms unstable nuclei into more stable nuclei through emission of alpha particles, beta particles, positrons and gamma radiation. The activity therefore

decreases exponentially with time, and graphical treatment of the data can be used to determine the half-life and decay constant.

The physicist warned that a weak source may fail to sterilise instruments adequately, while improper storage or disposal may expose workers, patients and nearby communities to harmful ionising radiation. The activity data below were recorded at equal time intervals.

Time / days	0	4	8	12	16	20	24	28	32	36	40
Activity	800	640	512	410	328	262	210	168	134	107	86

A decay constant greater than 10^{-3} s^{-1} represents a highly radioactive isotope.

Tasks

a) Explain the emissions that cause the activity of the source to decrease with time.

b) Analyse the data provided and use it to comment on the rate of decay of the radioactive nuclei in the hospital source.

c) Evaluate the health and environmental implications of using the radioactive source, and recommend measures the hospital should implement to minimise the risks.

4.

The Government of Uganda is promoting industrialisation to increase local production of ceramics, glass, electronic components and water-treatment chemicals. A consortium of manufacturers plans to establish an integrated industrial park near a limestone deposit and will use oxides obtained from different elements.

Before construction begins, the environmental and standards authorities require evidence that the selected oxides are suitable for their intended applications and that health and environmental risks have been considered. The compounds differ greatly in melting point, reaction with water and behaviour towards acids and alkalis.

The following data were provided.

Oxide	**Na_2O**	**MgO**	**Al_2O_3**	**SiO_2**	**CO_2**	**SO_3**
Melting point / °C	1132	2852	2072	1710	-78	17

Tasks

a) Analyze the information provided and use it to explain the differences observed among the oxides.

b) Relate the observed chemical behaviour of the oxides to their suitability for the proposed industrial applications.

c) Evaluate the health and environmental implications associated with the production and use of the oxides, and recommend appropriate measures for sustainable industrial development.

PART II

Respond to any **one** item in this part.

5.

A company that manufactures alcohols and polymer-based packaging materials has recently expanded its production. Its production units use hydrocarbons obtained from petroleum as raw materials for manufacturing alcohols and polymers.

Because of shortages of conventional raw materials, the company has been advised that methyl 2-methylpropenoate, a raw material for Perspex, and ethene, the monomer for poly(ethene), can each be prepared through synthetic routes beginning with propanone. The technical team also wants to understand the reactions by which poly(ethene) is formed from ethene and 1-bromopropan-2-ol is formed from propene so that likely process hazards can be controlled.

The company has been required to manage plastic waste effectively because poly(ethene) is non-biodegradable. It therefore needs a scientific evaluation of the material, its uses and appropriate waste-management options. Your chemistry teacher has provided you with the company brief as a professional problem-solving exercise.

Tasks

(a) Analyse the proposals concerning the preparation of the required raw materials from propanone.

(b) Explain the reactions involved in the formation of the alcohol and polymer products described in the scenario.

(c) Evaluate the use of the polymer product and propose appropriate measures for managing its waste.

6.

A small beverage and flavouring company in Namanve Industrial Park produces fruit-flavoured concentrates for local juice processors. One product is a pineapple-like ester prepared by heating propanoic acid with ethanol in the presence of concentrated sulphuric acid. The second is a bright yellow-orange solid formed when propanone reacts with acidified 2,4-dinitrophenylhydrazine.

Because of the attractive colour, some members of the production team have suggested adding the yellow-orange solid to drinks as a cheap food colour. The company has also received a large supply of ethanol from local distillers and wants to reduce production

costs by converting it into other useful raw materials, particularly propanone and propanoic acid, through suitable organic reaction pathways.

The company has invited Senior Six chemistry learners to prepare a scientific write-up that explains the chemistry, evaluates whether the proposed processes are realistic, and guides safe decisions before any product is released to consumers.

Tasks

a) Analyze the structures of the company's two key raw materials, other than ethanol, and help the production team understand how their functional groups account for differences in their physical properties.

b) Help the chemists understand the organic reaction mechanisms leading to formation of the final products and propose suitable pathways for obtaining useful raw materials from the locally supplied ethanol.

c) Advise the management on the possible impacts of using the organic compounds mentioned in the scenario and the precautions required before any product is released for human consumption.

END